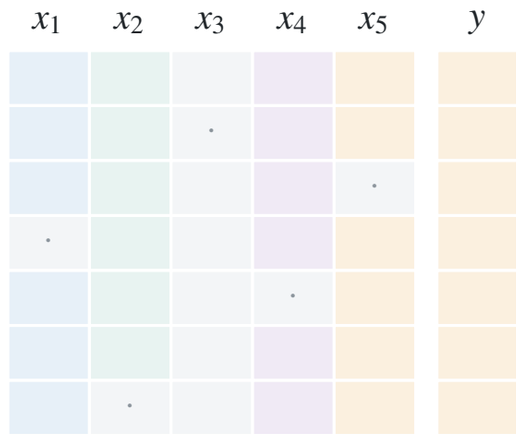
(a) Support and query as cells



grey: missing cell; sensor 3 is absent from the query

(b) Cell token transformer

attention across the cells of each row

attention across the rows of each column

$\times L$ blocks; query rows attend to support rows only

value or missing embedding + column embedding

(c) Three heads on the query row

Lifted site head (each observed query cell j)

corrections δ_j^{off}, δ_j^{slope}, δ_j^{D}, s_j

to the anchor site $\Rightarrow$ (Λ_j, η_j), zero at initialization

Free head (target token)

$p_{\text{free}}(y)$: Gaussian or three component mixture

Gate (target token)

$\pi = \sigma(g)$, gate bias +4 at initialization

(d) Anchor fitted on the support

incomplete support only

EM with missing entries, then factor analysis over $z = (y, h)$

$$x_j = w_j^\top z + c_j + e_j, \quad e_j \sim \mathcal{N}(0, \psi_j)$$

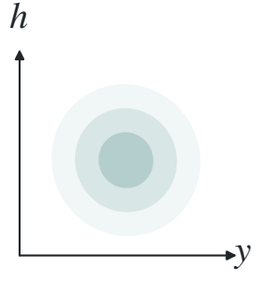
scale free: target standardized by the support mean and deviation

(e) Exact fusion of lifted sites

$$\Lambda = \Lambda_0 + \sum_{j \in S} \Lambda_j$$

$$\eta = \eta_0 + \sum_{j \in S} \eta_j$$

$$q(z) = \mathcal{N}(\Lambda^{-1}\eta, \Lambda^{-1})$$



a missing sensor adds no factor

marginal $p_{\text{site}}(y)$

(f) Linear opinion pool

$$p_{\text{UPT}}(y) = \pi p_{\text{site}}(y) + (1 - \pi) p_{\text{free}}(y)$$

$$-\log p_{\text{UPT}} \le \min\{-\log p_{\text{site}} - \log \pi, \; -\log p_{\text{free}} - \log(1 - \pi)\}$$

